

MathGov 15-File Core Audit Response and Finalization Report

Release: MathGov Core Release 2026.09 v12.3.1 / SGP v8.2.1

Date: 19 July 2026

Status: Final bounded public research-source and implementation-integrity patch

Scope: The selected 15-file review bundle and the complete supporting release package

Executive verdict

READY FOR PUBLIC AND ACADEMIC RESEARCH-SOURCE RELEASE, TEACHING, STRUCTURED TIER 1-2 USE, TIER 3 PROCEDURAL PILOTS, AND EMPIRICAL VALIDATION PREPARATION.

NOT CERTIFIED FOR HIGH-STAKES DEPLOYMENT, LEGAL COMPLIANCE, PHYSICAL SAFETY, CONSCIOUSNESS DETECTION, OR CLAIMS OF EMPIRICAL SUPERIORITY.

The framework architecture did not require redesign. The audit found that the core cascade, no-rescue rules, SGP type separations, WDBIP placement, consequence-tempo boundary, and non-decisive RLS behavior were substantively sound. The material defects were release-engineering, exact-version synchronization, workbook formula/interface integrity, and public-facing metadata. Those defects have been corrected in the v12.3.1 / SGP v8.2.1 patch line.

Two headline findings in the supplied external audit were not reproducible against the actual attached ZIP and were therefore rejected rather than implemented:

1. The active DOCX files in the attached ZIP are genuine OOXML Word documents, not Markdown files renamed with .docx extensions.
2. MATHGOV_REPRODUCIBILITY_AND_USE_STANDARD_v1.1.docx is present in docs/implementation/ and is included in the final 15-file review bundle.

The genuine workbook and synchronization findings were independently reproduced and corrected.

1. Audit method

The finalization pass applied MathGov's own Reality Grounding discipline to the package:

1. inspected the supplied ZIP rather than relying on filename claims;
2. classified active DOCX and XLSX files structurally;
3. compared filenames, internal titles, current control surfaces, manifests, and component pins;
4. traced the canonical cascade and component-ownership rules across documents;
5. inspected the Aligners workbook through its formula, validation, named-range, and current-release surfaces;
6. tested the run schema, state registry, transition matrix, semantic validator, WDBIP validator, positive vectors, and negative vectors;
7. regenerated and visually inspected changed document mirrors;
8. rebuilt the 15-file review bundle, release manifest, verification scripts, and hash ledger last.

A passing release verifier is treated as evidence of structural and procedural conformance only. It is not treated as proof that evidence inputs are true, thresholds are empirically calibrated, or decisions are morally or legally correct.

2. Finding disposition

2.1 Rejected findings

External finding	Final disposition	Basis
All selected DOCX files are plain Markdown carrying false .docx extensions	Rejected	Direct ZIP and OOXML inspection found word/document.xml, Word relationships, content types, styles, and valid ZIP containers in the active DOCX artifacts.
Reproducibility and Use Standard v1.1 is absent	Rejected	The exact DOCX, Markdown, and PDF are present under docs/implementation/ and are now copied into core_15/.
The release contains only 14 selected	Rejected for the actual attached ZIP	The final selected bundle contains 14

External finding	Final disposition	Basis
artifacts PC-AEP, MFDI, and Source Coupling all require forced content-version increments	Rejected	DOCX files and one XLSX workbook. Their substantive rules did not change. They remain v2.0, with current package placement and hashes. Mechanical version inflation would weaken semantic version discipline.
Password-protect every workbook formula as the primary integrity boundary	Reframed	Weak spreadsheet passwords are not a security boundary and conflict with open-source maintainability. Formula signatures, exact pins, named ranges, and load-bearing surfaces are now checked by the release verifier; an Edit Integrity Map identifies input, formula, fixed, review, and historical surfaces.

2.2 Accepted and corrected findings

Finding	Correction
Workbook split-brain current identity	Current control surfaces now pin Canon v12.3.1, SGP v8.2.1, ripple.md v5.2.1, Agent v12.1.1, Primer v4.1.1, WDBIP v1.3.1, and Aligners v5.3.1. Historical sheets remain explicitly historical.
Fragile ISFORMULA self-check	The workbook now performs portable YES/NO token-integrity checking. The external release verifier checks that every live flag cell remains formula-driven.
Sanity summary could omit errors	Summary now separately counts PASS, FAIL, formula errors, manual-review states, and unclassified nonblank statuses. Any FAIL, error, or unclassified state causes failure.
Off-by-one named ranges	CANON_SCHEMA_FAIL_COUNT now resolves to CANON!B115; CANON_PLACEHOLDER_COUNT resolves to CANON!B116.
“Computed” publish gates contained hardcoded attestations	Workbook-local publishability is formula-driven. Externally verified schema and package claims are clearly described as release-verifier responsibilities rather than internal workbook proof.
CVaR outputs were typed values	Visible Option A and Option B CVaR totals, scores, and pass states are now calculated from the displayed sorted scenario tables.
Dashboard omitted or hardcoded stage results	RG, RF/NCRC, TRC, CSV, and RLS results are linked to their live calculation or record surfaces. Level 5 is now explicit, including the refusal of deterministic selection.
Canon/SGP/ripple/Agent/WDBIP/RLS Validation/Public Intro/Primer pin drift	Exact component patch versions and filenames have been synchronized in current source surfaces and release metadata.
ripple.md companion matrix duplicated entries and named a nonexistent WDBIP version	The matrix is deduplicated and pins WDBIP v1.3.1. Wrapper feasibility tests are distinguished from the four qualification gates and RLS ranking layer.
WDBIP lacked a proper Markdown title surface	WDBIP now has a valid H1 and exact v1.3.1 patch identity.
Public Intro used an old core-release claim boundary	Corrected to v12.3.1 / SGP v8.2.1.
Agent Card and schema examples used old package pins	Corrected to v12.1.1 and the current Canon/SGP/wrapper package line.
RLS Validation vector headings used an old release label	Corrected to the current package line without changing the study design.
Primer used ambiguous shorthand for Level 4	First-use teaching language now gives the full CSV name while preserving accessible shorthand afterward.

3. Seven-C final scorecard

Criterion	Final score	Final assessment
Clear	9.1	Public doorway, compact cascade, component roles, WDBIP placement, SGP boundaries, and workbook workflow are explicit.

Criterion	Final score	Final assessment
Calculable	8.8	Dense specialist sections remain necessarily technical. RLS, NCRC, CVaR, state transitions, and worked-run formulas are operational. Empirical calibration remains provisional and is not confused with calculability.
Correct	9.0	No unresolved gate-order, formula-normalization, rights-rescue, or SGP type-collapse defect was found. Load-bearing workbook formulas and names are verifier-checked.
Consistent	9.2	Current package pins, filenames, component matrix, selected bundle, and workbook control surfaces are synchronized. Historical records remain clearly historical.
Coherent	9.3	The Canon controls decisions; SGP controls moral-status interfaces; WDBIP controls welfare measurement beneath RLS; ripple.md controls portable assurance; Agent System controls runtime implementation.
Complete	9.0	The selected 15 files cover the declared core review purpose, while the full package supplies schemas, validators, state matrices, examples, tests, manifests, and release support.
Ready	8.7	Ready for research-source publication and bounded structured use. Not ready in the sense of empirical validation or certified consequential deployment, which the release expressly refuses to claim.

Supplementary assessment:

- scientific claim discipline: 9.3;
- falsifiability architecture: 9.0;
- rights integrity: 9.4;
- catastrophic-risk integrity: 9.1;
- CSV structural-viability integrity: 9.2;
- SGP type integrity: 9.3;
- WDBIP measurement-boundary integrity: 9.2;
- machine and spreadsheet integrity: 9.0;
- procedural reproducibility: 8.9;
- empirical maturity: 3.0, correctly disclosed as provisional.

4. Architecture verdict

The canonical architecture remains:

RG -> RF/NCRC -> TRC -> CSV -> RLS

The patch does not add a sixth gate. It does not turn WDBIP into an option gate. It does not turn UCI or HOI into headline stages. It does not permit RLS to compensate for rights, ruin, or structural failure.

The five levels have distinct functions:

1. **Reality Grounding:** establishes claim boundaries, evidence traces, unknowns, category grounding, causal admissibility, and refusal conditions.
2. **Rights Floor / NCRC:** tests non-compensatory rights constraints.

3. **TRC:** tests catastrophic and irreversible tail-risk boundaries.
4. **CSV:** tests containment, structural viability, execution controls, source coupling, physical admissibility, dependency, reversibility, monitoring, and authority conditions.
5. **RLS:** ranks only qualified and selectable survivors.

This order is not merely explanatory. It prevents attractive aggregate benefits from laundering deeper violations.

5. File-by-file final verdict

5.1 RippleLogic Canon v12.3.1

Role: principal governing source for decision architecture.

Verdict: ready as the normative semantic source within the declared Tier 1-3 research architecture.

The patch corrects exact SGP, WDBIP, wrapper, Agent, validation, and public-package pins. It adds no new gate or equation. The Canon remains extensive, but the compact Reproducibility Standard and Cascade Standard reduce the implementation burden.

5.2 Sentience Gradient Protocol v8.2.1

Role: principal governing source for moral-patienthood evidence, protection, participation, power-readiness, intelligence, and reality-management capacity.

Verdict: ready as a research and governance protocol, not a consciousness detector.

The patch preserves:

- Human FPP-100;
- the current known humanity RMCP-P100 calibration anchor;
- the open plateau across biological, digital, hybrid, collective, and possible extraterrestrial intelligence;
- no P101 or superior caste;
- capacity-performance separation;
- capacity-authority separation;
- capacity-standing separation;
- RMCP misuse review, suspension, and retirement conditions.

5.3 ripple.md v5.2.1

Role: portable assurance wrapper and standard.

Verdict: ready as a portable assurance specification.

The consequence-tempo rule correctly compares the worst credible control critical path against an unacceptable adverse escalation or irreversible stop-loss boundary, not against the first intended effect. Responsibility continuity preserves answerability across human, model, vendor, committee, and operational handoffs. The patch only repairs package matrix and terminology drift.

5.4 Agent System v12.1.1

Role: runtime and deployment-control companion.

Verdict: ready for implementation research and controlled pilots, not certified deployment.

The current Agent Card, operator-kit, and schema examples now use the active package pins. Selection, authorization, and execution remain separate. Runtime timing, intervention authority, emergency logging, and responsibility chains are explicit.

5.5 CSV Gate Standard v2.1

Role: compact CSV status and routing standard.

Verdict: ready without a content version increment.

It correctly treats CSV as a viability and containment gate rather than a purity test. The status ladder preserves PASS, PASS_WITH_CONTROLS, REDESIGN_REQUIRED, ESCALATE, FAIL, EMERGENCY_PROVISIONAL, and NOT_MATERIAL distinctions.

5.6 Cascade Standard v2.3

Role: compact authoritative implementation map.

Verdict: ready without a content version increment.

Its strongest functions are gate order, no-rescue, short-circuit states, emergency separation, confidence discipline, selectability, RLS normalization, and refusal of deterministic selection.

5.7 Reproducibility and Use Standard v1.1

Role: minimum replayable operating kernel and proportional-use standard.

Verdict: present, complete for its declared purpose, and ready.

It defines Quick, Standard, and Audit profiles; parameter status; record requirements; state transitions; replay difference classes; and claim boundaries. It does not claim that reproducible procedure proves correct evidence or moral truth.

5.8 WDBIP v1.3.1

Role: Canon-subordinate normative implementation companion beneath RLS.

Verdict: correctly included in the selected core review set.

WDBIP is important enough to include because the RLS field cannot be reliably populated without rules preventing dimension collapse, dimension isolation, duplicate effects, scope inflation, subgroup masking, time-window mismatch, dependence laundering, and unsupported causal arrows. It is not a third principal source, new gate, or eighth dimension.

5.9 RLS Validation Protocol v2.3.1

Role: validation-study and falsification protocol.

Verdict: ready to govern Level 1 validation research.

It correctly narrows early claims to comprehension, disagreement, burden, fatigue, missingness, preliminary reliability, and boundary confusion. Strong latent-structure or validation claims require later preregistered sample justification and evidence.

5.10 Foundations Primer v4.1.1

Role: informative human doorway.

Verdict: ready.

It preserves the three-pillar and two-phase teaching structure without replacing the formal five-stage method. It explains WDBIP, tempo, responsibility, SGP, and physical-admissibility boundaries without turning them into extra public steps.

5.11 Public Introduction v12.3.1

Role: shortest public teaching surface.

Verdict: ready.

Its central teaching form remains:

- Reality, Rights, Ripples;
- first qualify;
- then rank.

It correctly refuses legal, empirical, and deployment claims.

5.12 PC-AEP v2.0

Role: physical and causal admissibility evidence profile.

Verdict: ready; no semantic increment warranted.

It prevents governance permission, model fluency, or formal compliance from being treated as proof that a physical intervention is safe or causally realizable.

5.13 MFDI v2.0

Role: methodological falsifiability and dependency-integrity standard.

Verdict: ready; no semantic increment warranted.

It supports claim typing, dependency tracing, alternative explanations, revision triggers, and re-derivation after foundational changes.

5.14 Source Coupling Integrity Standard v2.0

Role: source-use and dependency-integrity standard.

Verdict: ready; no semantic increment warranted.

It separates source availability, source quality, source coupling, source debt, and downstream claim authority.

5.15 Aligners Sheet v5.3.1

Role: bounded Tier 2 worked-run exemplar, training surface, and audit aid.

Verdict: ready in that bounded role; it is not a validator.

The final workbook now provides:

- exact active release pins;
- formula-driven stage results;
- a visible five-level dashboard;
- formula-derived CVaR results from displayed scenario tables;
- complete PASS/FAIL/ERROR/unclassified sanity accounting;
- corrected machine named ranges;
- formula-driven workbook-local publishability;
- release-verifier formula-presence checks;
- WDBIP, consequence-tempo, responsibility, reproducibility, and workflow surfaces;
- an open-source Edit Integrity Map.

Historical sheets remain available for lineage but do not control the current release.

6. Calculability and measurement boundary

MathGov distinguishes five statuses that are commonly collapsed:

1. **Mathematically defined:** the formula or predicate is formally stated.
2. **Computationally implemented:** the result can be calculated from supplied inputs.
3. **Procedurally reproducible:** qualified implementers can replay the stated process or classify their divergence.
4. **Empirically calibrated:** the inputs, thresholds, or instrument have supporting validation evidence.
5. **Normatively legitimate:** the governance commitment or authority is justified within the applicable constitutional and legal context.

The release has strong status at levels 1-3 for many components. It does not claim universal achievement of levels 4-5.

The following remain empirical or governance research questions:

- RLS construct validity and calibration;
- dimension-boundary reliability;
- cross-cultural measurement invariance;
- UCI and HOI predictive value;
- kernel propagation validity;
- MPS threshold calibration;
- RMCP cross-domain and cross-substrate validity;
- P100 calibration stability;
- domain-specific rights and TRC thresholds;
- CSV measurement instruments;
- consequence-tempo threshold calibration.

7. Machine and reproducibility status

The package contains:

- JSON Schema Draft 2020-12 run records;
- a validator separating structural and semantic checks;
- a canonical state registry;
- a state-transition matrix;
- positive and intentionally failing run vectors;
- a WDBIP schema and semantic validator;
- positive and intentionally failing WDBIP vectors;
- exact current-pin verification;
- workbook formula/interface verification;
- document and PDF mirror checks;

- a complete hash ledger.

Passing these checks establishes only that the supplied records and artifacts satisfy the implemented conformance rules. It does not establish that an evidence source is true, a causal model is valid, a right is correctly interpreted under law, or a deployment is safe.

8. Readiness by use case

Use	Final readiness
Public explanation	Ready
Academic working-paper and research-source review	Ready
Open-source framework publication	Ready
Classroom teaching	Ready
Tier 1 structured use	Ready within declared limits
Tier 2 structured use	Ready within declared limits
Tier 3 procedural pilots and independent replay	Ready with qualified review and domain evidence
Empirical validation studies	Ready to begin; results not yet established
Organizational deployment	Requires domain-specific evidence, authority, monitoring, and controls
High-stakes public/autonomous deployment	Not certified by this release

9. Remaining limitations and future research

No further architecture file is required before release. The highest-value next work is empirical rather than architectural:

1. freeze a case packet and conduct independent two-implementer replay;
2. run the Level 1 RLS/WDBIP comprehension and reliability study;
3. conduct consequence-tempo pilots across fast protective action, rapid automation, and slow institutional lock-in;
4. test responsibility-continuity records against real multi-vendor and human-model handoffs;
5. publish negative, null, mixed, and failure findings;
6. revise or retire constructs that do not add decision value.

10. Final claim boundary

The v12.3.1 / SGP v8.2.1 package establishes a synchronized, internally checked, rights-constrained, tail-risk-sensitive, structurally governed, welfare-measurement-aware, assurance-ready decision architecture for research, teaching, structured runs, replay, and validation preparation.

It does not establish:

- empirical truth of all inputs;
- universally valid welfare dimensions or weights;
- legal or constitutional authority;
- physical or medical safety;
- consciousness or personhood detection;
- metaphysical proof;
- superior decisions across domains;
- ProofPack or Tier 4 completion;
- production deployment certification;
- automated moral truth.

The appropriate final description is:

MathGov v12.3.1 / SGP v8.2.1 is architecture-complete and release-integrity ready for its declared public research-source, teaching, structured-run, replay, and validation-preparation purposes. Its empirical instruments and real-world performance remain open to testing, challenge, revision, narrowing, and retirement.